

Attachment E — Figures

Each figure states, under its title, the custody arrangement, the number of children, whether child care is included in the order, and the incomes used. The text below says what each one shows and what it does not.

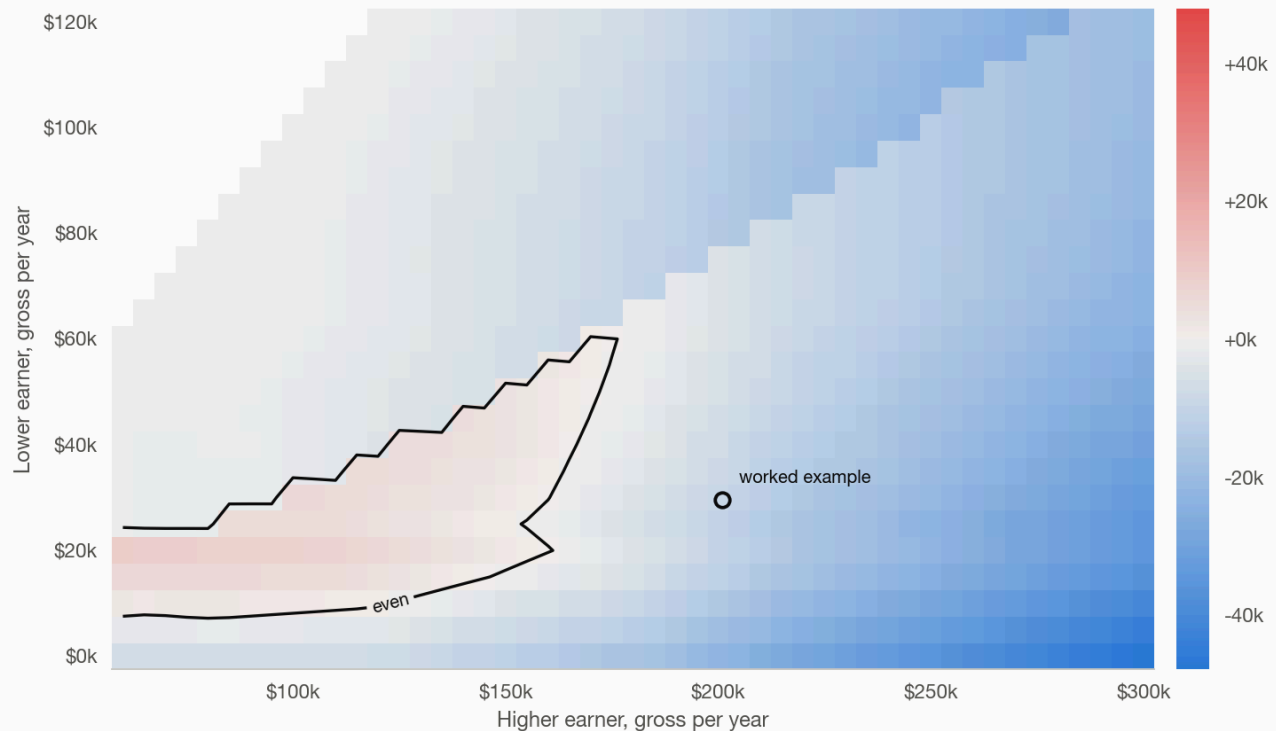
Who holds more after the order

E01. With three children, the recipient household holds more after the order in 10 percent of income combinations. Three children, joint custody, no child care. The colour is the recipient household's net income minus the payor's, per year, across every combination of a higher earner from \$60,000 to \$300,000 and a lower earner from \$0 to \$120,000. The recipient household is ahead in 75 percent of the grid below \$150,000 of higher-earner income and nowhere above \$175,000. The closed contour inside the red region is not noise: it is Line 6e's limitation ceasing to bind once the payor's Line 6d crosses 10 percent, at which point the order drops by about \$110 a week in a single step.

With three children, the recipient household holds more after the order in 10% of income combinations

Recipient household net minus payor net, per year.

CUSTODY Joint, equal time (Box 1) · CHILDREN 3 · CHILD CARE None (base support) · INCOMES Vary (axes)



Notes: Measured from the grid's own cells (fig1_heatmap_3child_box1.csv). The inner 'even' loop is Line 6e's limit ending at 6d = 10%: the order drops ~\$110/wk in one step. Masked where the lower earner would out-earn the higher. Premiums \$43/\$33 a week; MA under-13 credit set to zero.

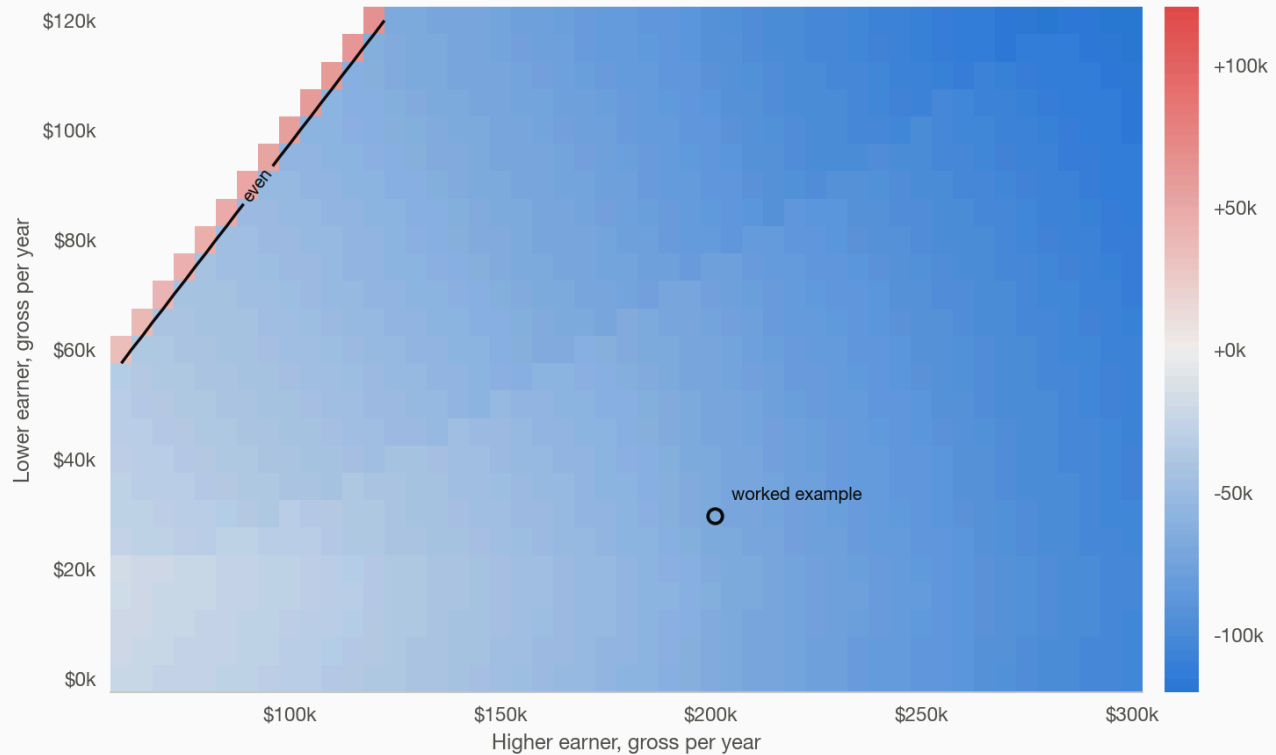
Source: model/worksheet.py (CJ-D 304, matches the form's own scripts); model/net_position.py (TY2026 federal + MA tax)

E02. Per person, the payor holds more almost everywhere. The same grid divided by household size, one against four. The payor is ahead in 99 percent of the grid. This figure travels with E01 wherever E01 is shown, because the per-person comparison is the one a reader will raise first.

Per person, the payor holds more almost everywhere

Recipient household net $\div 4$, minus payor net. Companion to E01.

CUSTODY Joint, equal time (Box 1) · CHILDREN 3 · CHILD CARE None (base support) · INCOMES Vary (axes)



Notes: Payor ahead in 99% of cells. Masked where the lower earner would out-earn the higher. Premiums \$43/\$33 a week; MA under-13 credit set to zero.

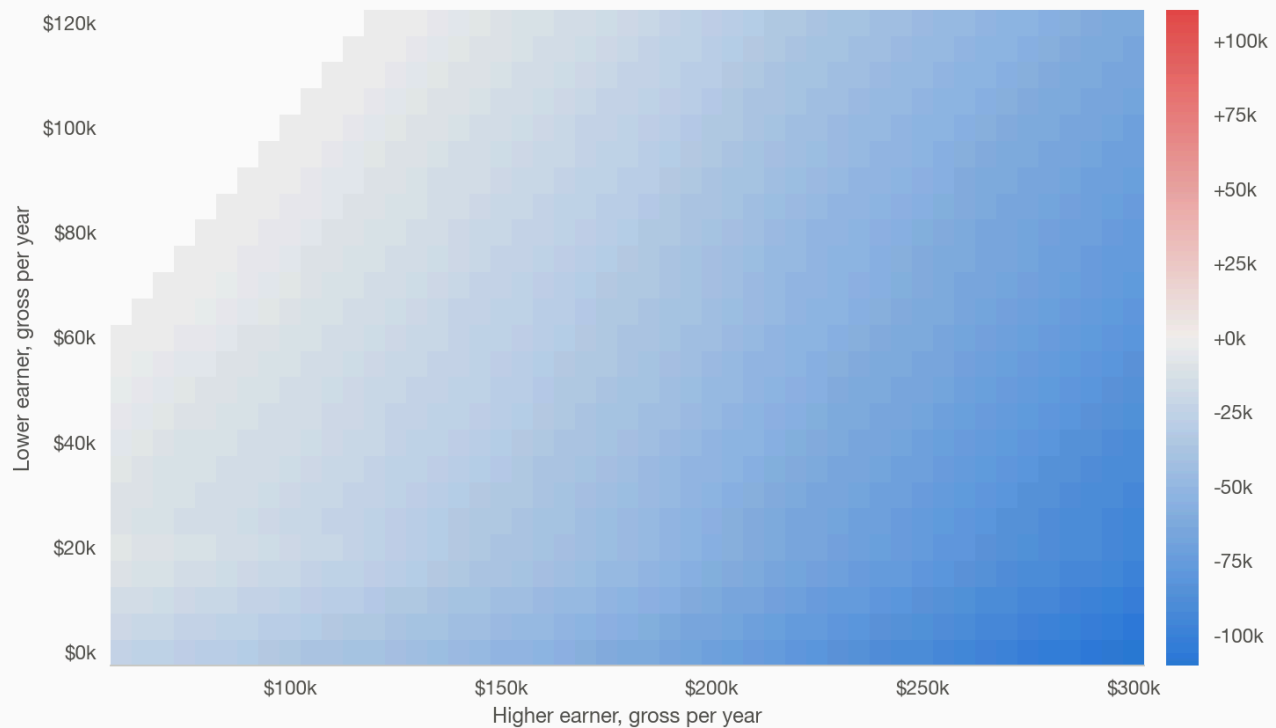
Source: model/worksheet.py (CJ-D 304, matches the form's own scripts); model/net_position.py (TY2026 federal + MA tax)

E18. With one child, the recipient household holds more after the order in 0 percent of income combinations. The same grid and colour scale as E01, one child instead of three. Companion to E01 and E19: the number of children, not any one family's facts, decides which household comes out ahead. (Previously the first panel of a three-panel exhibit; split out under this project's one-exhibit-one-graph rule.)

With one child, the recipient household holds more after the order in only 0% of income combinations

Recipient household net minus payor net, per year. Companion to E01/E19 (same grid, 2 and 3 children).

CUSTODY Joint, equal time (Box 1) · CHILDREN 1 · CHILD CARE None (base support) · INCOMES Vary (axes)



Notes: Measured from the grid's own cells (fig1_heatmap_1child_box1.csv). Masked where the lower earner would out-earn the higher. Premiums \$43/\$33 a week; MA under-13 credit set to zero.

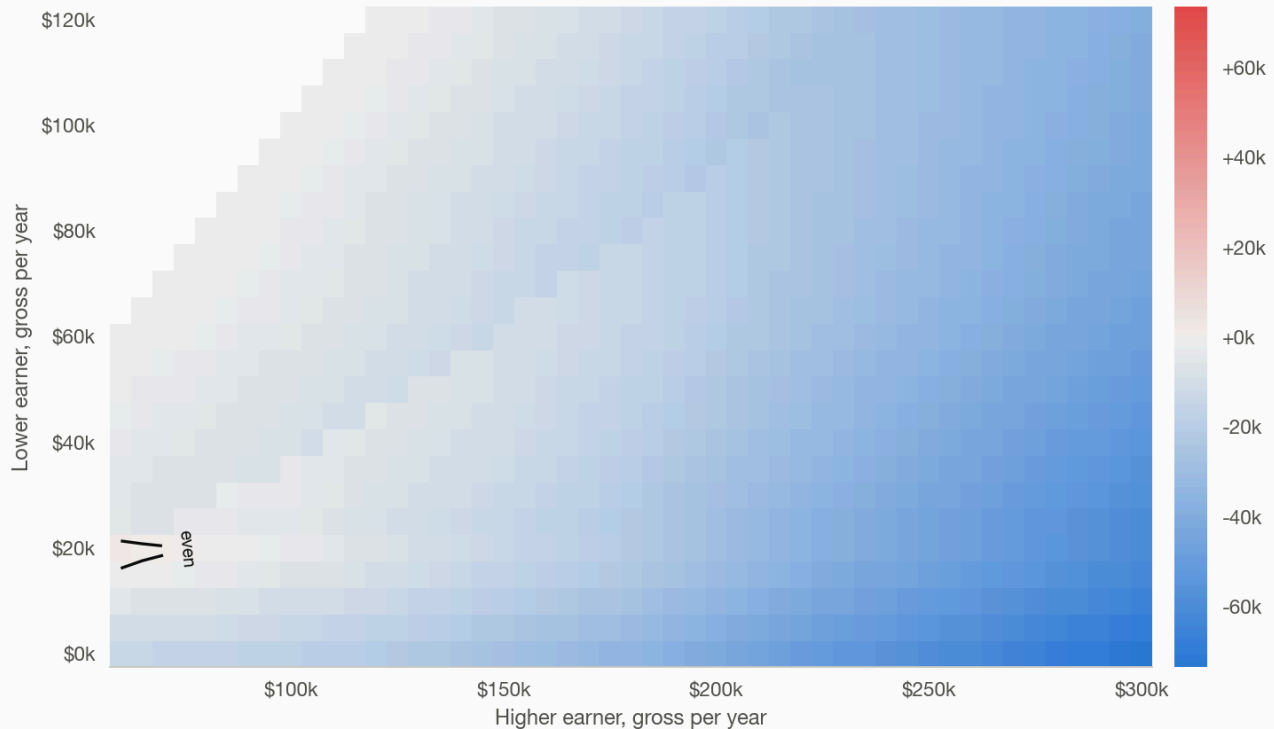
Source: model/worksheet.py (CJ-D 304, matches the form's own scripts); model/net_position.py (TY2026 federal + MA tax)

E19. With two children, the recipient household holds more after the order in 0.3 percent of income combinations (4 of 1,147). The same grid and colour scale as E01 and E18, two children. Companion to both. Unlike E18's one child, this is not a genuine zero, just a small one: four cells out of 1,147, all at a lower earner of \$20,000 with the higher earner between \$60,000 and \$75,000, the bottom corner of the grid. (Previously the second panel of the same three-panel exhibit.)

With two children, the recipient household holds more after the order in 0% of income combinations

Recipient household net minus payor net, per year. Companion to E01/E18 (same grid, 1 and 3 children).

CUSTODY Joint, equal time (Box 1) · CHILDREN 2 · CHILD CARE None (base support) · INCOMES Vary (axes)



Notes: Measured from the grid's own cells (fig1_heatmap_2child_box1.csv). Masked where the lower earner would out-earn the higher. Premiums \$43/\$33 a week; MA under-13 credit set to zero.

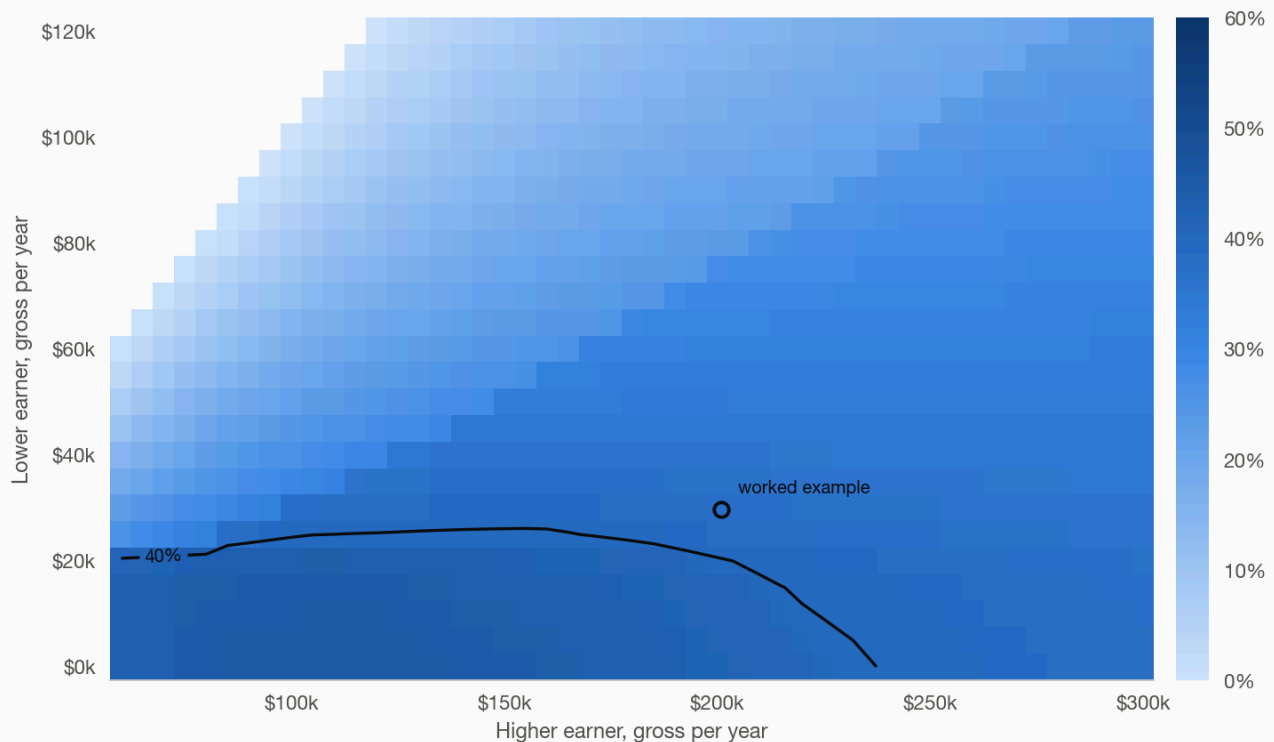
Source: model/worksheet.py (CJ-D 304, matches the form's own scripts); model/net_position.py (TY2026 federal + MA tax)

E04. The order exceeds 40 percent of the payor's net income only where the lower earner makes about \$25,000 or less. The 40 percent contour, computed on net income, sits at a lower-earner income of about \$25,000 up to roughly \$235,000 of higher-earner income; above that, the order never reaches 40 percent of net at any lower-earner income on the grid. The region under the contour is the one Section IV.C's hardship presumption is written for, and E05 shows what the Worksheet reports there.

The order exceeds 40% of the payor's net income only where the lower earner makes about \$20,000 or less

Order as a share of payor net; line at 40%.

CUSTODY Joint, equal time (Box 1) · CHILDREN 3 · CHILD CARE None (base support) · INCOMES Vary (axes)



Notes: 15% of cells. Scale fixed 0–60%. Masked where the lower earner would out-earn the higher. Premiums \$43/\$33 a week; MA under-13 credit set to zero.

Source: model/worksheet.py (CJ-D 304, matches the form's own scripts); model/net_position.py (TY2026 federal + MA tax)

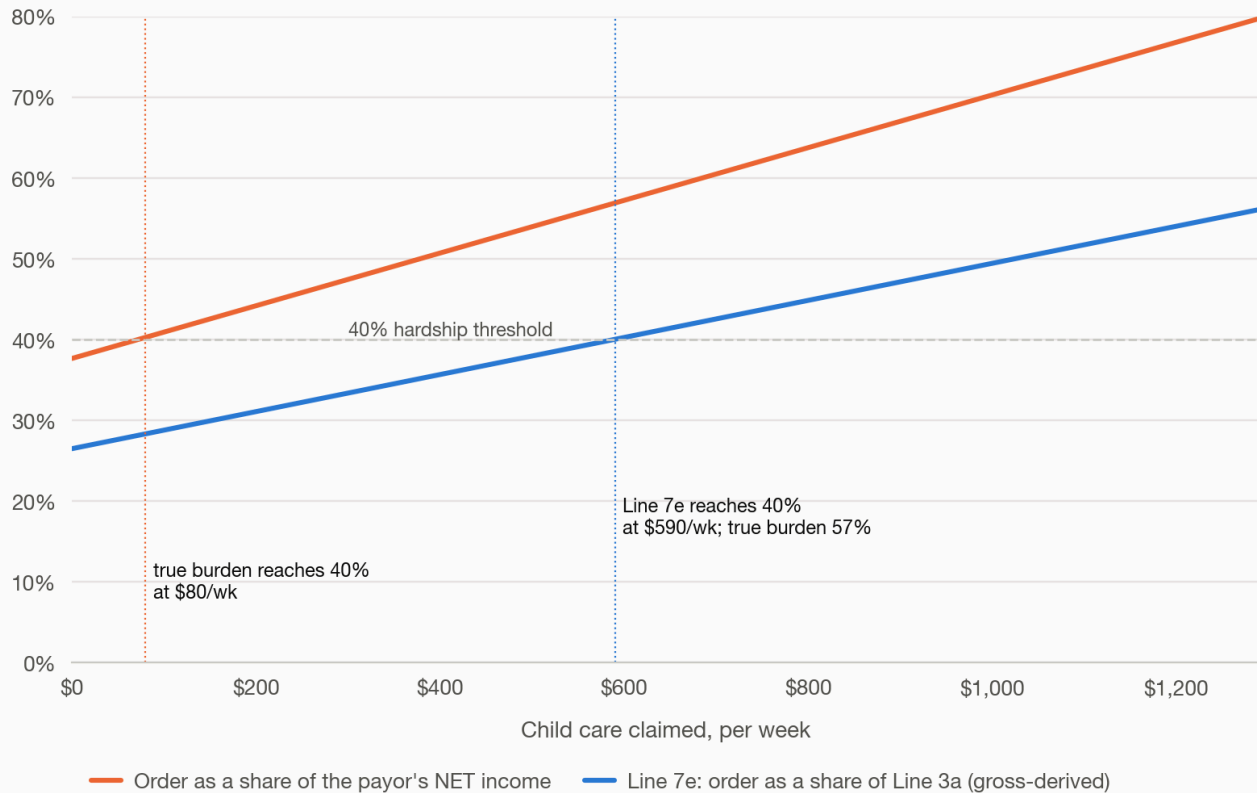
The hardship test and its units

E05. The hardship valve fires late because it reads the wrong income. At the worked example, Line 7e divides the order by gross-derived available income while the order is paid from net income. As child care claimed by the recipient rises, the true burden passes 40 percent of net at \$80 a week of child care; Line 7e reports 40 percent at \$590 a week, by which point the true burden is 57 percent. Nothing on the form flags the gap.

The hardship test fires late because it reads the wrong income

Line 7e's reading vs the true share of net income as claimed child care rises.

CUSTODY Joint, equal time (Box 1) · CHILDREN 3 · CHILD CARE Rises to the \$430/child ceiling (recipient)
INCOMES \$201,000 / \$29,640



Notes: Child care is included in the order. 7e is what the form computes (Guidelines Section IV.C); net is what is paid. Two of three children under 13 (MA credit); premiums \$43/\$33.

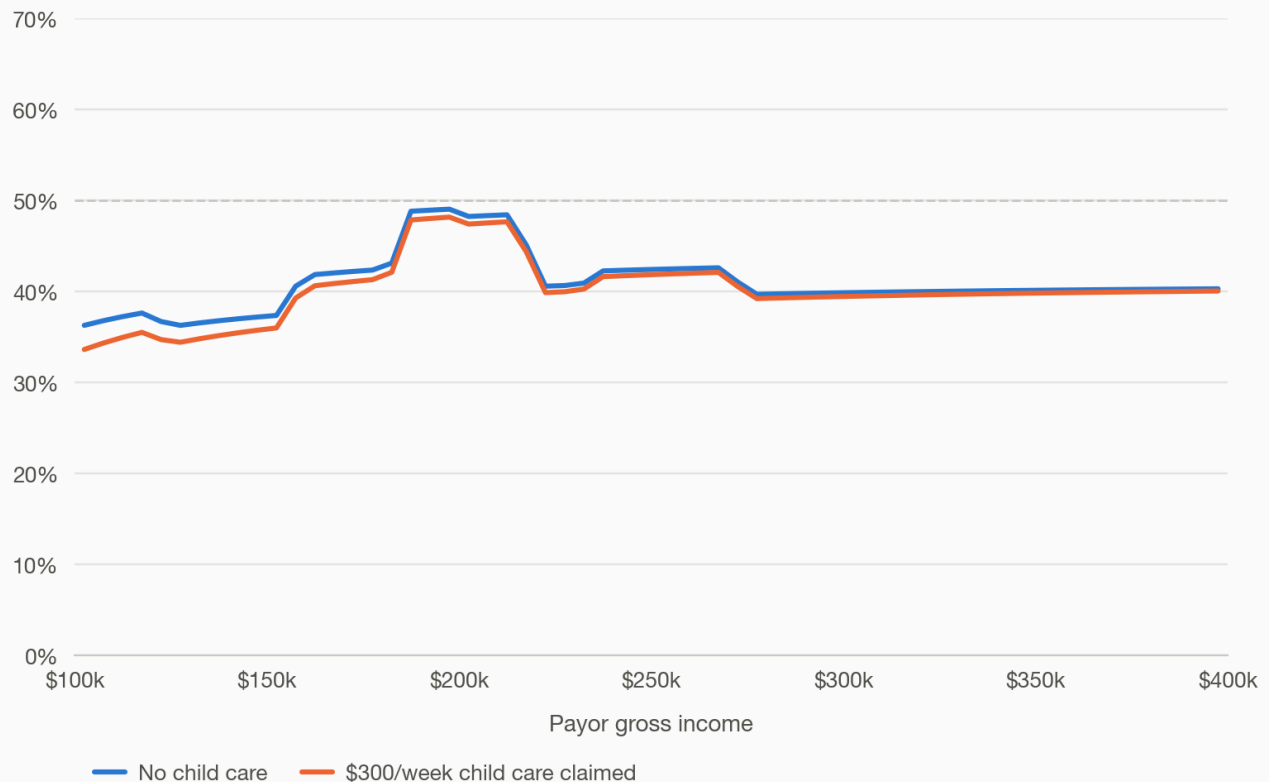
Source: model/submission_figures.py

E10. Of the payor's next dollar, the payor keeps between a third and a half. Marginal retention after federal and state tax and the resulting change in the order, from \$100,000 to \$400,000 of payor income, with and without \$300 a week of child care. The order's marginal take runs near 20 percent because the child-count multiplier and the rising income share both scale with the payor's income, so the effective rate exceeds Table A's 10 percent top bracket.

Of the payor's next dollar, the payor keeps between a third and a half

After federal and state tax and the change in the order.

CUSTODY Joint, equal time (Box 1) · CHILDREN 3 · CHILD CARE None vs \$300/wk (recipient) · INCOMES Varies / \$29,640



Notes: \$5,000 steps. Dashed: 50 cents. Two of three children under 13 (MA credit); premiums \$43/\$33.
Source: model/marginal_retention.py

Child care

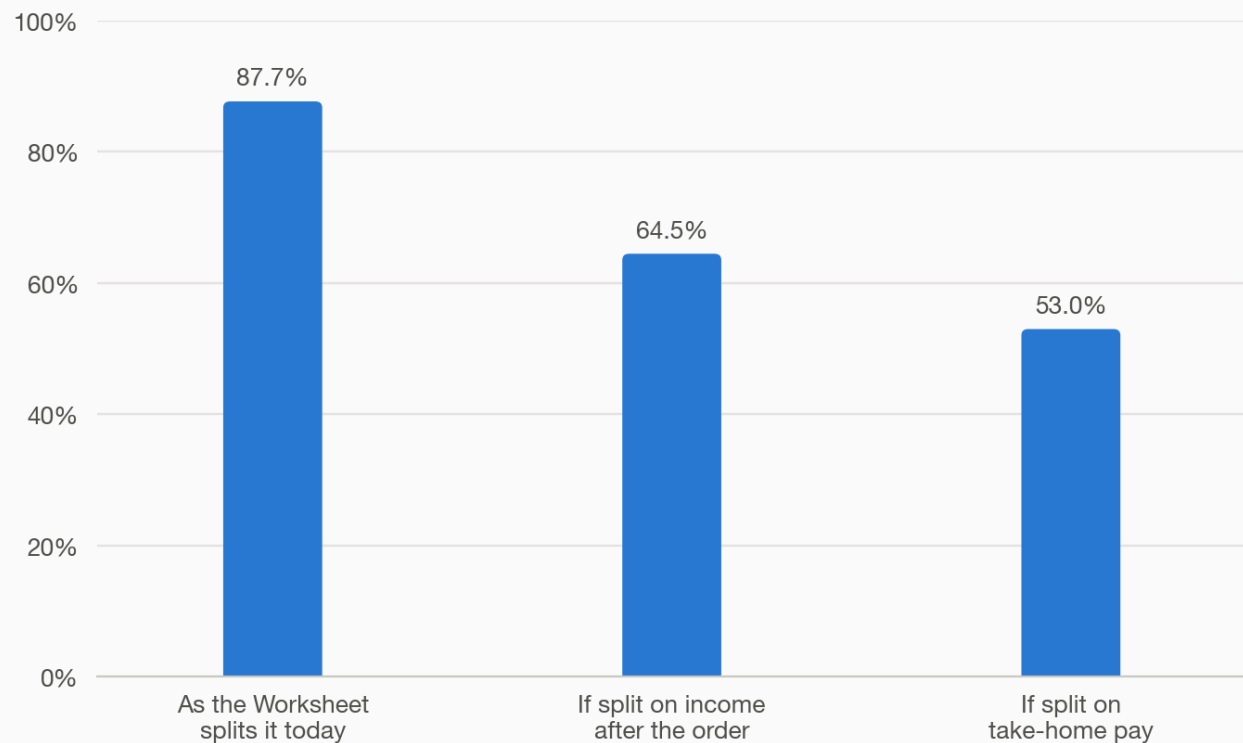
Eo6. The payor's share of a \$15,600 child care bill, four ways to split it. Line 3c allocates 87.7 percent to the payor on pre-transfer income shares. The letter's § 2 primary redline (new Lines 6b-1a and 6b-1) measures each parent's share of what is left after the order and after tax and FICA withholding, and brings it to 53.0 percent. § 2's fallback (new Line 6b-2), which needs no tax computation, brings it to 64.5 percent. Counting the federal and Massachusetts refundable credits on top of the primary redline's basis, under the alternating-year Child Tax Credit convention Box 1 now uses, brings it to 49.7 percent; that figure is analysis, not a request, because it turns on each parent's filing status and which parent claims which child, facts the Worksheet does not collect. The comments ask for the withholding-basis figure, not the credits-inclusive one.

The payor's share of a \$15,600 child care bill, three ways to split it

Payor's share of the same \$15,600 bill under three allocation rules.

CUSTODY Joint, equal time (Box 1) · CHILDREN 3 · CHILD CARE \$300/wk, paid by the recipient

INCOMES \$201,000 / \$29,640



Notes: Fallback is new Line 6b-2, gross basis; primary redline (the ask) is new Lines 6b-1a and 6b-1, withholding basis. Two of three children under 13; premiums \$43/\$33.

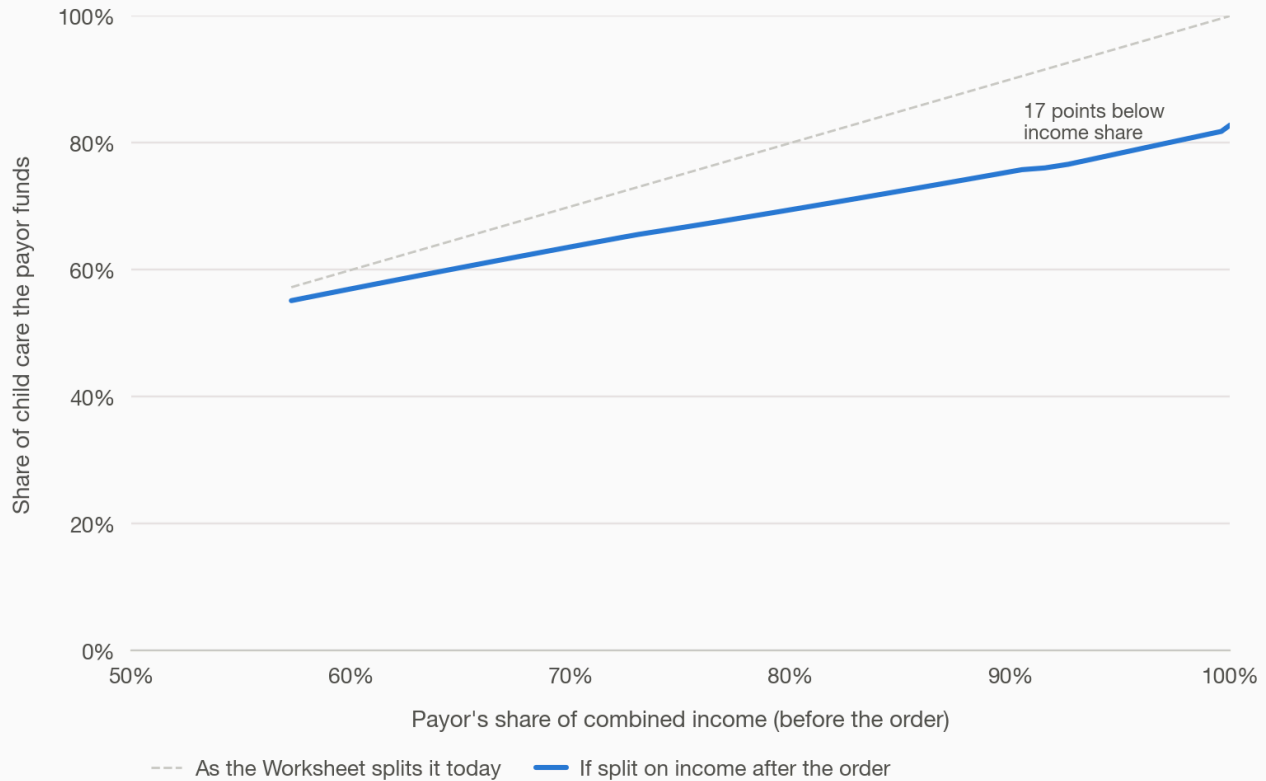
Source: model/childcare_post_transfer.py

E2o. With one child, post-transfer child care funds the payor 28 points below his income share. Share of the child care bill the payor funds against his pre-transfer income share (Line 3c), one child, \$100 a week paid by the lower earner, under two post-transfer rules; the gap is measured at the highest income share plotted. (Previously the first panel of a three-panel exhibit.)

With one child, splitting child care on income after the order funds the payor 17 points below his income share

Share of the child care bill the payor funds, by his income share before the order, under the current rule and one alternative.

CUSTODY Joint, equal time (Box 1) · CHILDREN 1 · CHILD CARE \$100/child/wk, paid by the lower earner
INCOMES \$201,000 / varies



Notes: Child care is included in the order. Gap measured at the highest income share plotted. Premiums \$43/\$33 a week; MA under-13 credit set to zero.

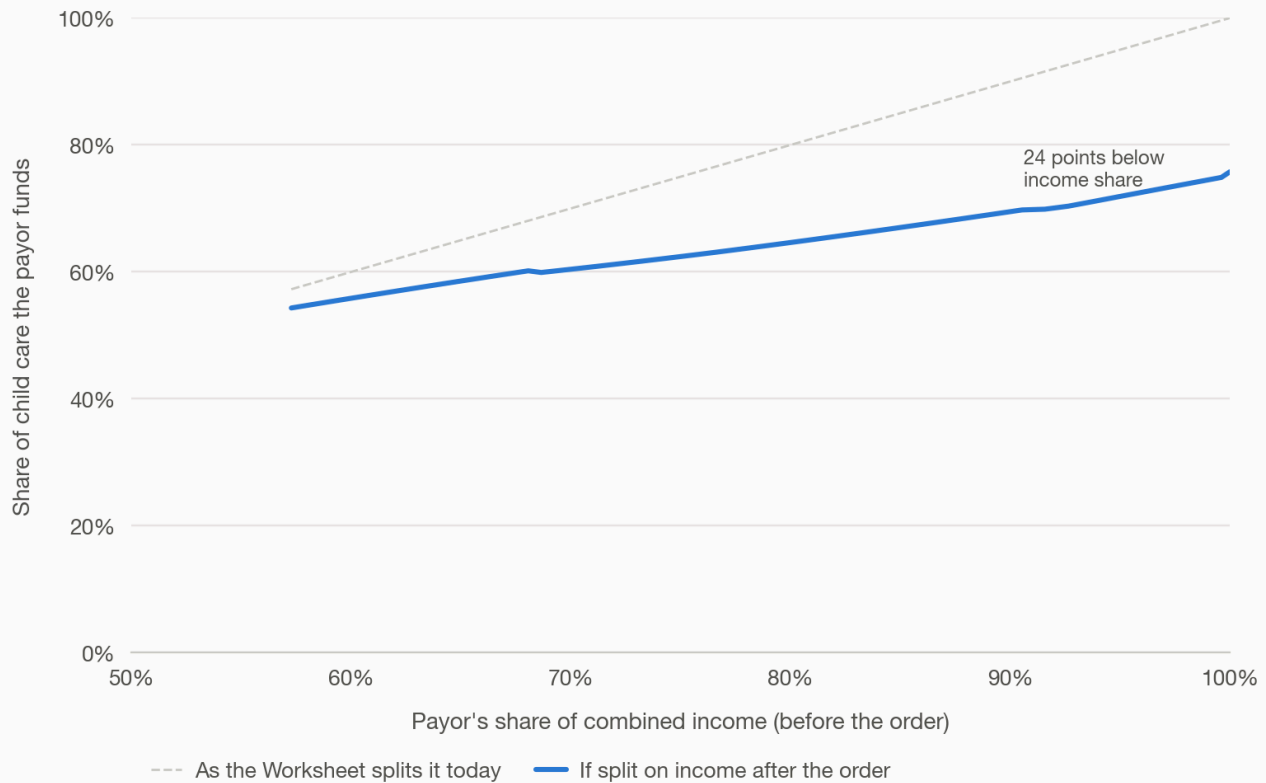
Source: model/worksheet.py (CJ-D 304, matches the form's own scripts); model/net_position.py (TY2026 federal + MA tax); fig2_childcare.py

E21. With two children, post-transfer child care funds the payor 39 points below his income share. Same construction as E20, two children. The gap widens with the number of children. (Previously the second panel of the same exhibit.)

With two children, splitting child care on income after the order funds the payor 24 points below his income share

Share of the child care bill the payor funds, by his income share before the order, under the current rule and one alternative.

CUSTODY Joint, equal time (Box 1) · CHILDREN 2 · CHILD CARE \$100/child/wk, paid by the lower earner
INCOMES \$201,000 / varies



Notes: Child care is included in the order. Gap measured at the highest income share plotted. Premiums \$43/\$33 a week; MA under-13 credit set to zero.

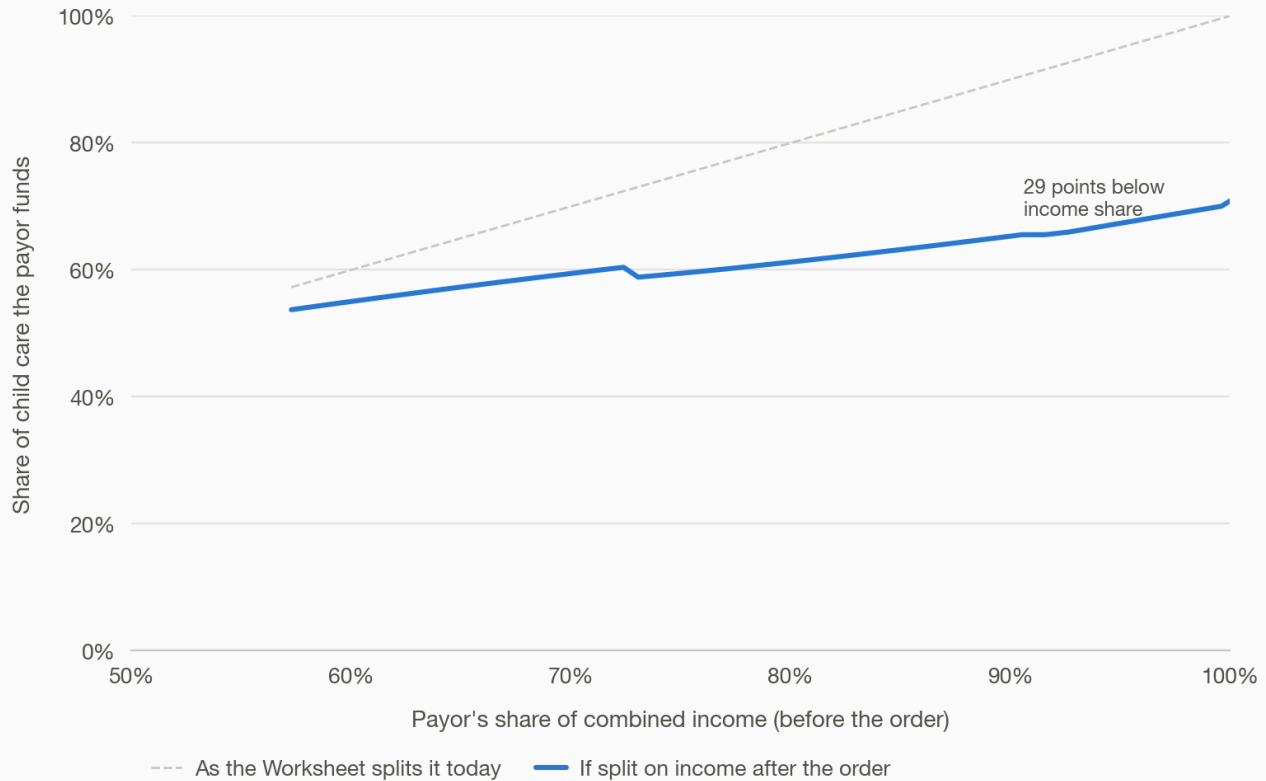
Source: model/worksheet.py (CJ-D 304, matches the form's own scripts); model/net_position.py (TY2026 federal + MA tax); fig2_childcare.py

E22. With three children, post-transfer child care funds the payor 46 points below his income share. Same construction as E20 and E21, three children — the worked example's own child count. (Previously the third panel of the same exhibit.)

With three children, splitting child care on income after the order funds the payor 29 points below his income share

Share of the child care bill the payor funds, by his income share before the order, under the current rule and one alternative.

CUSTODY Joint, equal time (Box 1) · CHILDREN 3 · CHILD CARE \$100/child/wk, paid by the lower earner
INCOMES \$201,000 / varies



Notes: Child care is included in the order. Gap measured at the highest income share plotted. Premiums \$43/\$33 a week; MA under-13 credit set to zero.

Source: model/worksheet.py (CJ-D 304, matches the form's own scripts); model/net_position.py (TY2026 federal + MA tax); fig2_childcare.py

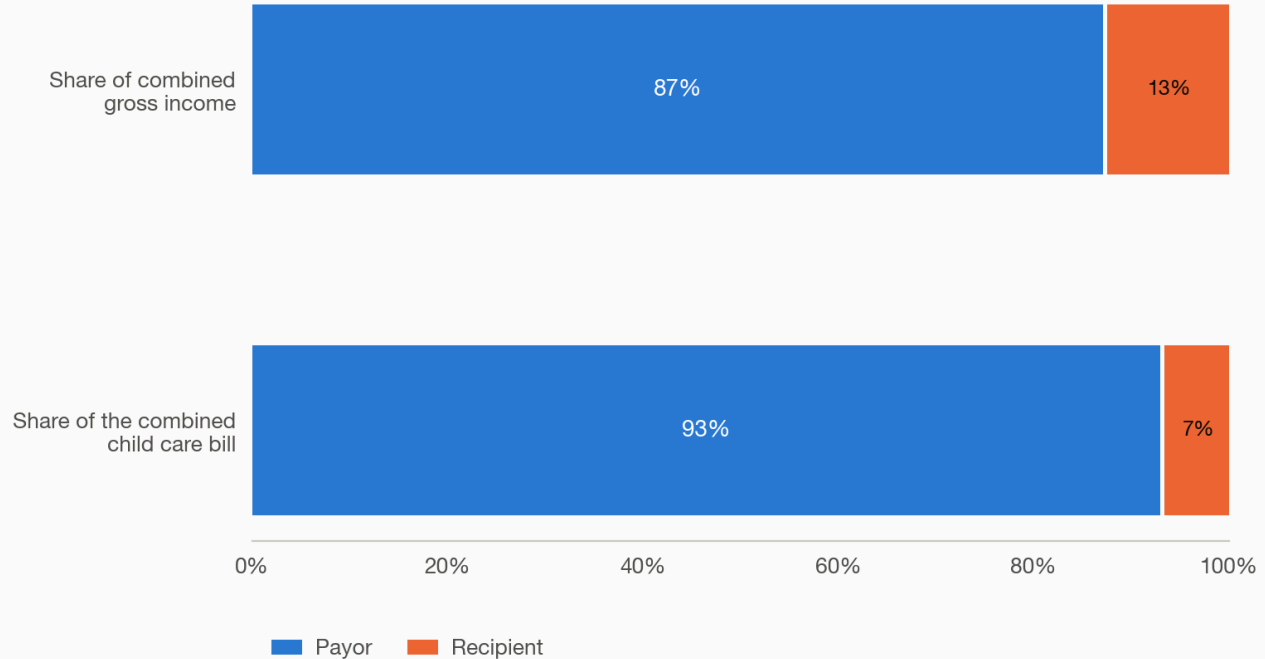
E25. The payor bears 93 percent of the combined child care while earning 87 percent of the gross income. Each parent pays \$300 a week during their own parenting time, the ordinary case at equal time. Two stacked bars: share of combined gross income and share of the combined \$31,200 child care bill. (Previously the first panel of a two-panel exhibit.)

The payor bears 93% of the combined child care while earning 87% of the gross income

Who pays a combined \$31,200/yr of child care, both parents paying \$300/wk in their own home.

CUSTODY Joint, equal time (Box 1) · CHILDREN 3 · CHILD CARE \$300/wk in each home

INCOMES \$201,000 / \$29,640



Notes: Companion to E26 (each parent's net position under three scenarios). Two of three children under 13.

Source: model/submission_figures.py § 2.1; fig8_both_pay.py

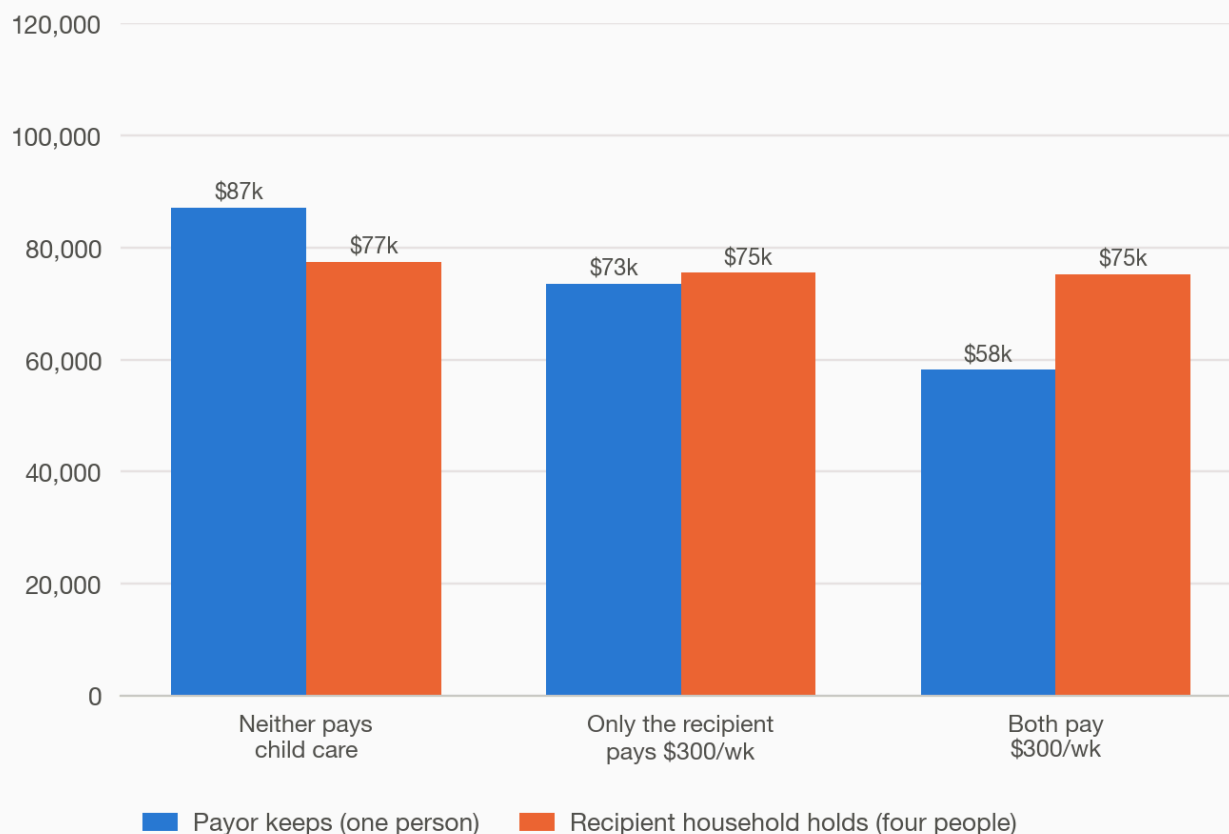
E26. Both parents paying child care costs the payor \$29,008 a year, the recipient household \$2,192. Each parent's net position under three scenarios — neither pays, only the recipient pays \$300 a week, both pay \$300 a week. The Worksheet allocates the recipient's cost to the payor through Line 6b and passes the payor's own cost back through Line 6e at about two cents on the dollar, so his own \$15,600 reduces the order by only \$270 a year. Order plus his own child care reaches 58 percent of his net income while Line 7e reads 33 percent. Per person the payor still leads. (Previously the second panel of the same exhibit.)

Both parents paying child care costs the payor \$29,008 a year, the recipient household \$2,192

Each parent's net position, neither, one, or both paying \$300/wk of child care.

CUSTODY Joint, equal time (Box 1) · CHILDREN 3 · CHILD CARE \$0 / \$300 / \$300 per wk (see scenarios)

INCOMES \$201,000 / \$29,640



Notes: Companion to E25. The payor's own \$15,600 cuts the order by only \$270/yr (Line 6e clip). Per person he still leads.

Source: model/submission_figures.py § 2.1; fig8_both_pay.py

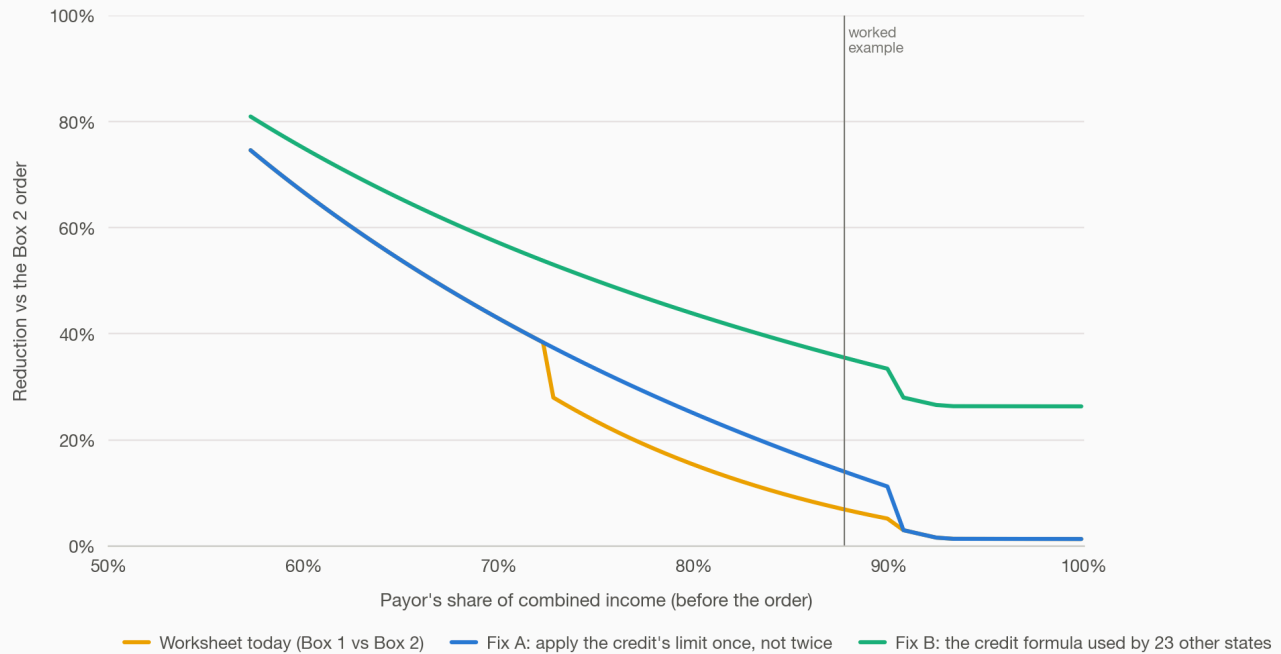
Parenting time

Eo8. The Worksheet's credit for equal parenting time collapses as the income gap widens; a cross-credit narrows without collapsing. The reduction in the order for equal time, measured against what the same parents would pay with the children a third of the time, falls from about 75 percent at a 57 percent payor income share to 7 percent at 88 percent and 1 percent at 96 percent, because no parenting-time quantity enters any line. Variant A, applying Line 6e once, lifts the curve; Variant B, the standard cross-credit at a 1.5 duplication factor used by 23 states, stops it collapsing.

The Worksheet's equal-time credit collapses as the income gap widens

Reduction in the order for equal time, against the Box 2 order (the paying parent has the children about a third of the time).

CUSTODY Joint (Box 1) vs primary (Box 2) · CHILDREN 3 · CHILD CARE None (base support) · INCOMES \$201,000 / varies



Notes: Fix A and Fix B are this project's proposed redlines (letter § 5; box1_fix.py). Premiums \$43/\$33 a week; MA under-13 credit set to zero.

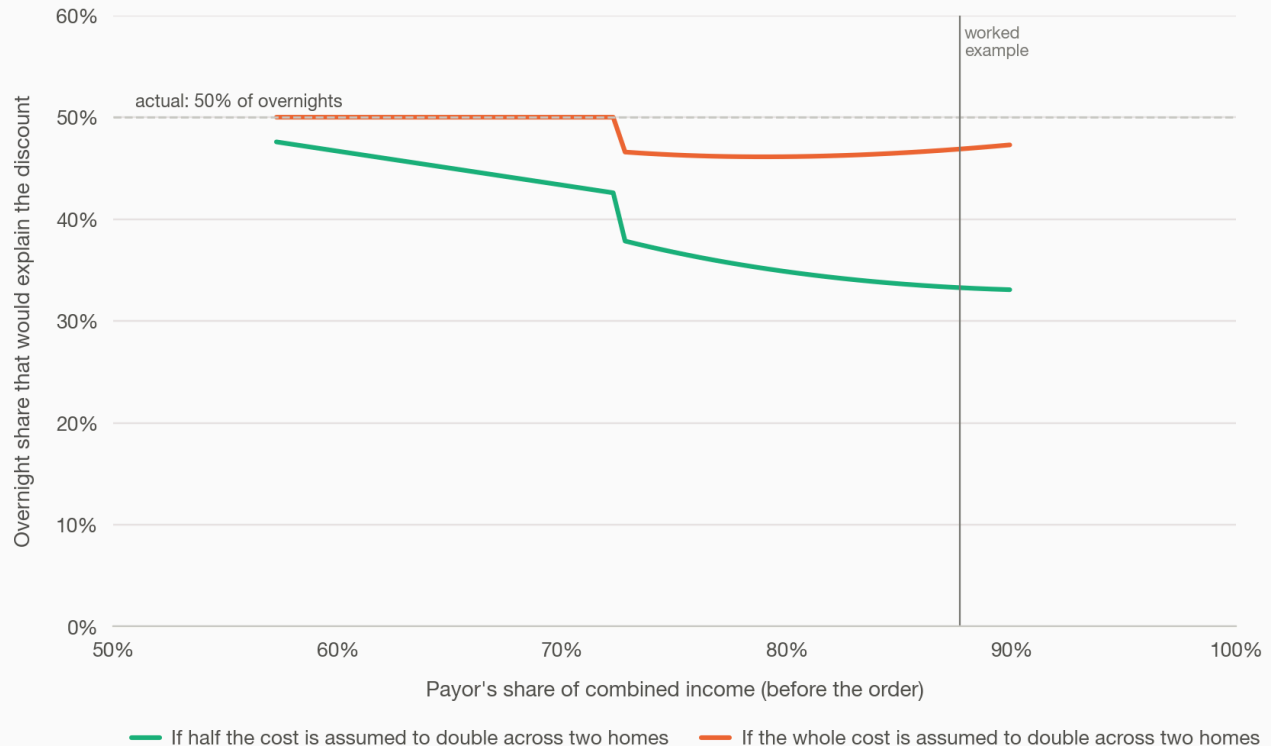
Source: model/worksheet.py (CJ-D 304, matches the form's own scripts); model/net_position.py (TY2026 federal + MA tax)

E09. At the worked example, equal parenting time is priced as a third of overnights at factor 1.5, and 47 percent at factor 2.0. The overnight share that reproduces the Box 1 order under a standard cross-credit. The number depends on the duplication factor, so the factor is always stated beside it. The curve is blank below the Line 5c floor, where the order is no longer a cross-credit.

The equal-time credit is what a standard formula pays a parent who has the children one night in three

Massachusetts's credit for half the nights, converted into the overnight share a standard formula would need to produce it.

CUSTODY Joint, equal time (Box 1) · CHILDREN 3 · CHILD CARE None (base support) · INCOMES \$201,000 / varies



Notes: "Double across two homes" is the duplication factor: 33% of overnights at 1.5 (Indiana's assumption), 47% at 2.0 (Box 1's own structure). Blank below the Line 5c floor. Premiums \$43/\$33 a week; MA under-13 credit set to zero.

Source: model/worksheet.py (CJ-D 304, matches the form's own scripts); model/net_position.py (TY2026 federal + MA tax); box1_fix.py

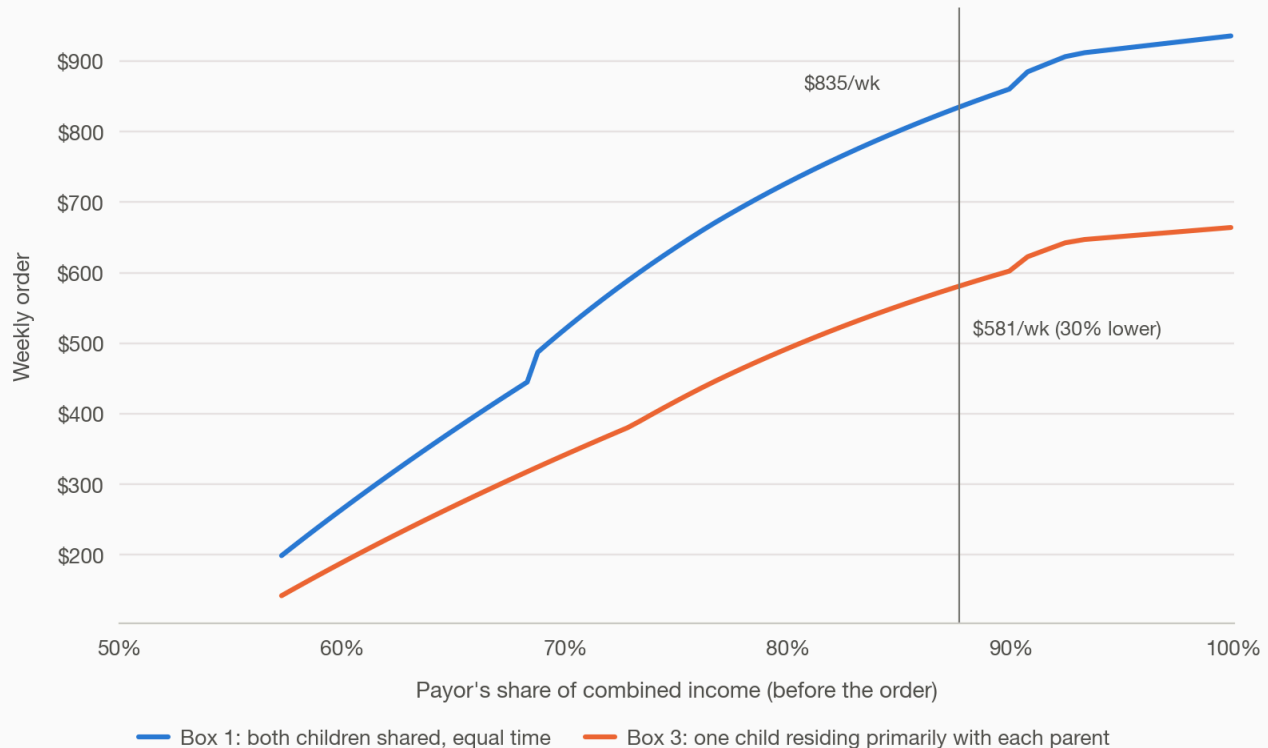
E23. Splitting two children across two homes cuts the weekly order by 30 percent.

Under Box 1 both children are shared equal time; under Box 3 one lives primarily with each parent. The payor's care responsibility is one child-share either way, but Line 6g nets the columns on the one-child schedule, so the Box 3 order comes out lower even though it costs more (E24). This one cuts against the lower earner. (Previously the first panel of a two-panel exhibit.)

Splitting two children across two homes cuts the weekly order by 30%

Weekly order for the same two children: both shared equal time (Box 1) vs one child with each parent (Box 3).

CUSTODY Joint (Box 1) vs split (Box 3) · CHILDREN 2 · CHILD CARE None (base support) · INCOMES \$201,000 / varies



Notes: Companion to E24 (the schedule's own cost moves the opposite way). Letter § 5.1. Premiums \$43/\$33 a week; MA under-13 credit set to zero.

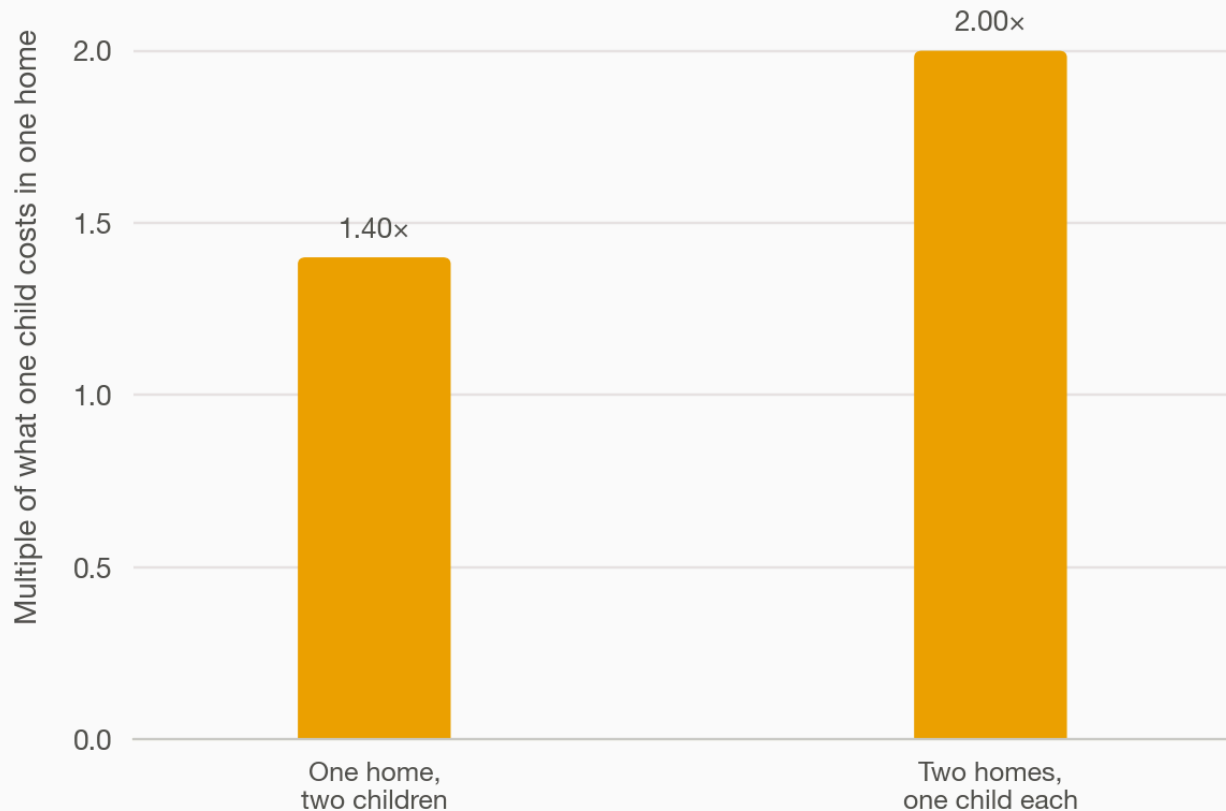
Source: model/worksheet.py (CJ-D 304, matches the form's own scripts); model/net_position.py (TY2026 federal + MA tax); fig7_box3_split.py

E24. Two homes, one child each, cost 43 percent more on the schedule than one home with two. Table B's own cost multiple: one home raising two children is charged 1.40 times the one-child amount; two homes each raising one child are charged 1.00 twice, for 2.00. The order itself moves the opposite way (E23). (Previously the second panel of the same exhibit.)

Two homes, one child each, cost 43% more on the schedule than one home with two

What the guidelines assume each arrangement costs: one home raising two children vs two homes each raising one.

CUSTODY Joint (Box 1) vs split (Box 3) · CHILDREN 2 · CHILD CARE None (base support)
INCOMES \$201,000 / varies



Notes: Companion to E23 (the order itself falls even as the schedule's own cost rises). Letter § 5.1.

Source: model/worksheet.py (CJ-D 304, matches the form's own scripts); model/net_position.py (TY2026 federal + MA tax); fig7_box3_split.py

Fifty-one jurisdictions

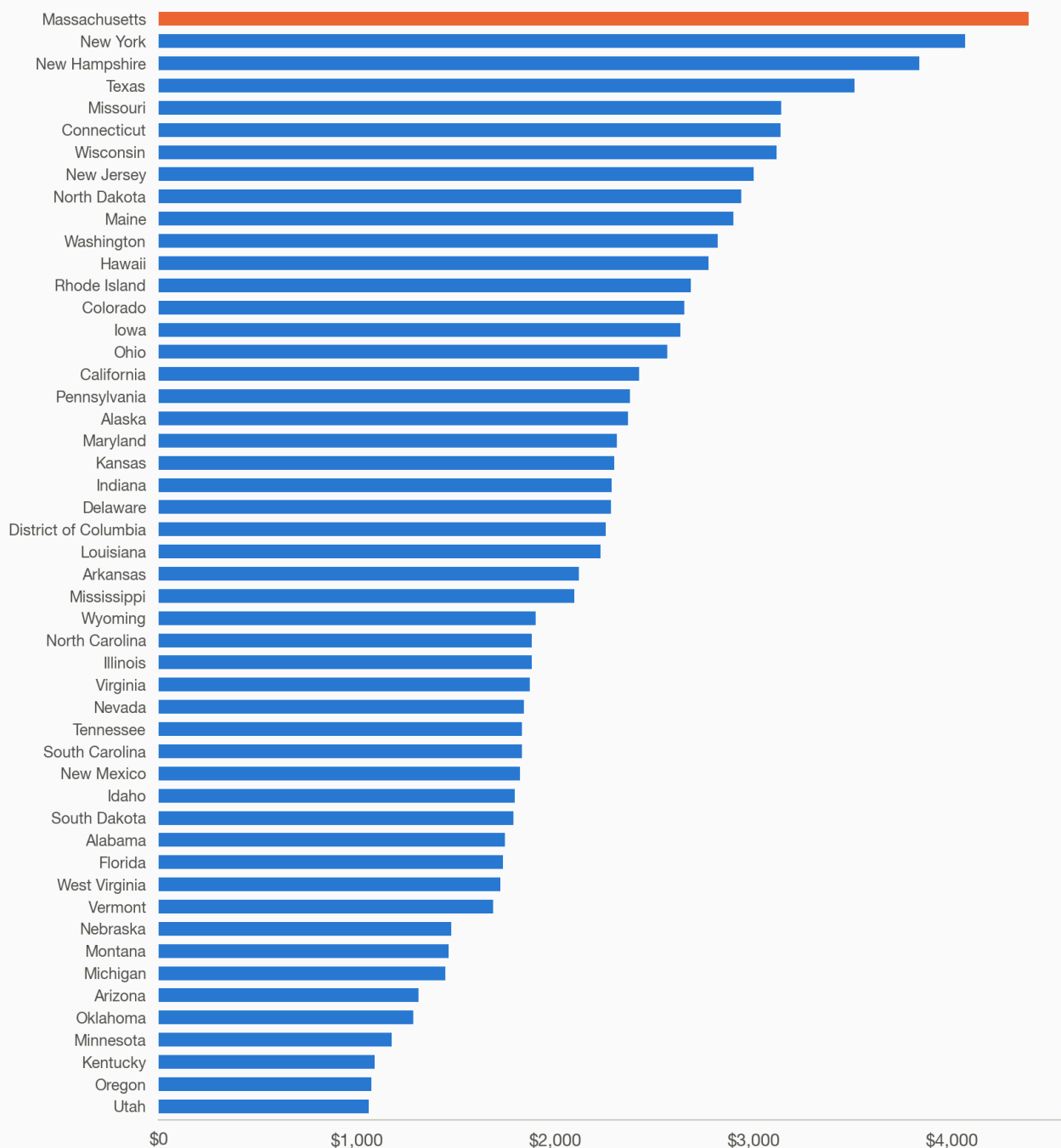
E11. Equal parenting time: Massachusetts orders the most of fifty jurisdictions.

Monthly order at one fact pattern, three children, \$201,000 and \$29,640, no child care. Fifty of fifty-one survived every verification stage; Georgia is held out because its enacted formula produces a lower order under primary custody than under equal time, which the state's own calculator reproduces. One fact pattern; the ranking generalises to nothing else.

At equal parenting time, Massachusetts orders the most of the fifty states

Monthly order at one fact pattern. Georgia held out.

CUSTODY Joint, equal time (Box 1) · CHILDREN 3 · CHILD CARE None (base support) · INCOMES \$201,000 / \$29,640



Notes: Each row profiled from primary sources, computed twice blind, reconciled, attacked. Own premiums (\$43/\$33) as each state treats them. One fact pattern only.

Source: data/fifty-state/tier-50-2026-09-05.json

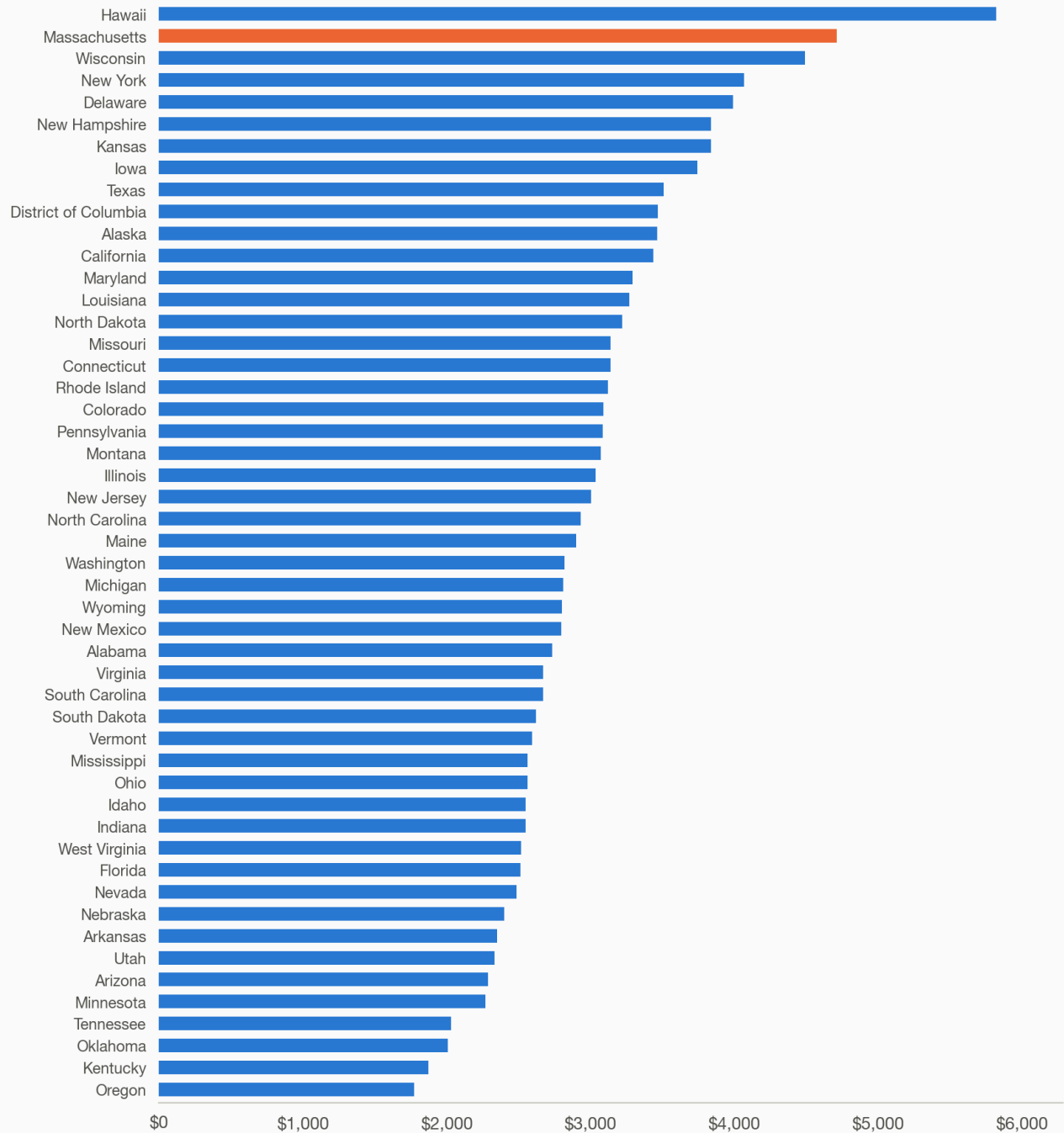
E12. Lower earner primary: only Hawaii's Melson formula orders more than Massachusetts. The same fact pattern with the children primarily with the lower earner. Hawaii's design is shared with Delaware and Montana; at this income gap it exhausts the self-support reserve differently from an income-shares schedule. First of a deliberate pair with E17: same states, same axis range, same colours, same ordering rule.

With the lower earner primary, only Hawaii orders more than Massachusetts

Monthly order at one fact pattern. Georgia held out.

CUSTODY Primary with the lower earner (Box 2) · CHILDREN 3 · CHILD CARE None (base support)

INCOMES \$201,000 / \$29,640



Notes: Each row profiled from primary sources, computed twice blind, reconciled, attacked. Own premiums (\$43/\$33) as each state treats them. One fact pattern only. Same states, axis, and colors as E17, so the two read as a pair.

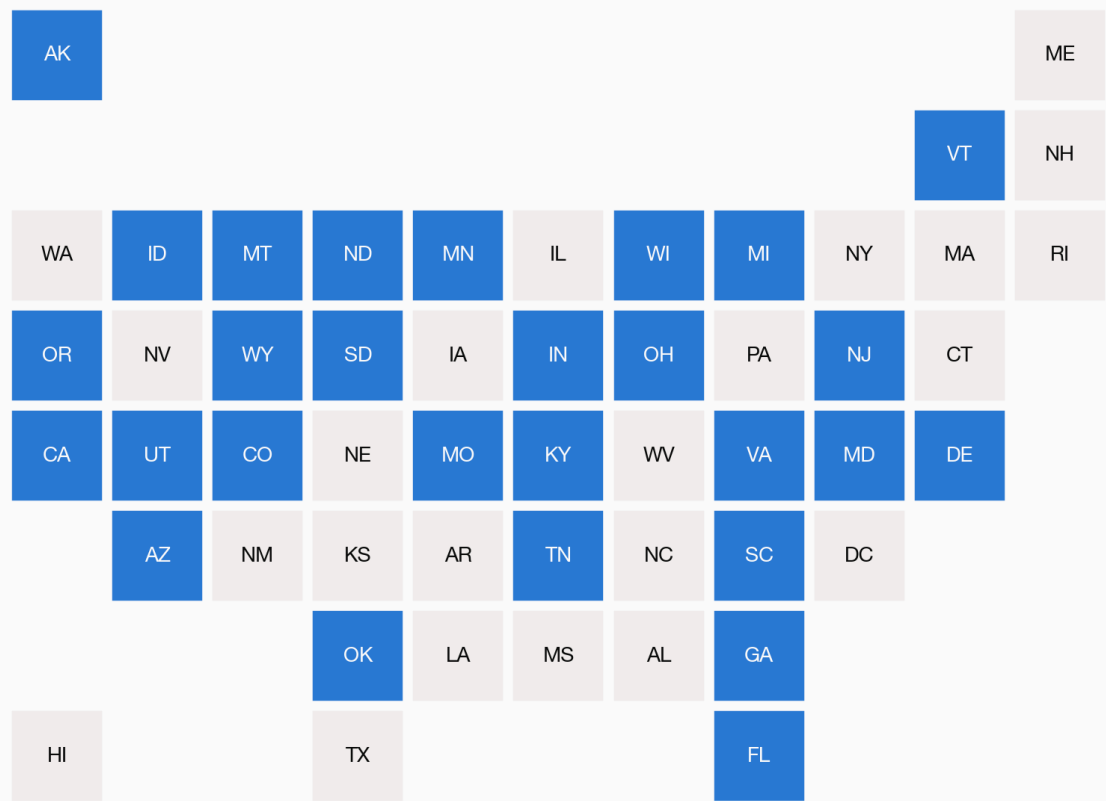
Source: data/fifty-state/tier-50-2026-09-05.json

E13. A parent with the children a third of the time gets a formula credit in 28 jurisdictions and none in 23. A count, not a dollar amount. Massachusetts is among the 23: its Box 2 is the only box below equal time, so the Worksheet produces the same number whether the paying parent has the children a third of the nights or none at all. In the one clean pairing with a

credit-giving state the dollar effect runs the other way, so the count is a structural fact and nothing more.

In Massachusetts and 22 other states, having the children a third of the time lowers the order by nothing
Blue: a formula credit at 122 overnights a year. Grey: none.

COUNTED Any formula credit at 122 overnights · CHILDREN 3 · CHILD CARE None · INCOMES \$201,000 / \$29,640



Notes: A count, not a dollar amount. Massachusetts's Box 2 already assumes a parent has the children about a third of the time, without a discrete credit for it.

Source: data/fifty-state/credit-at-122-2026-09-05.json

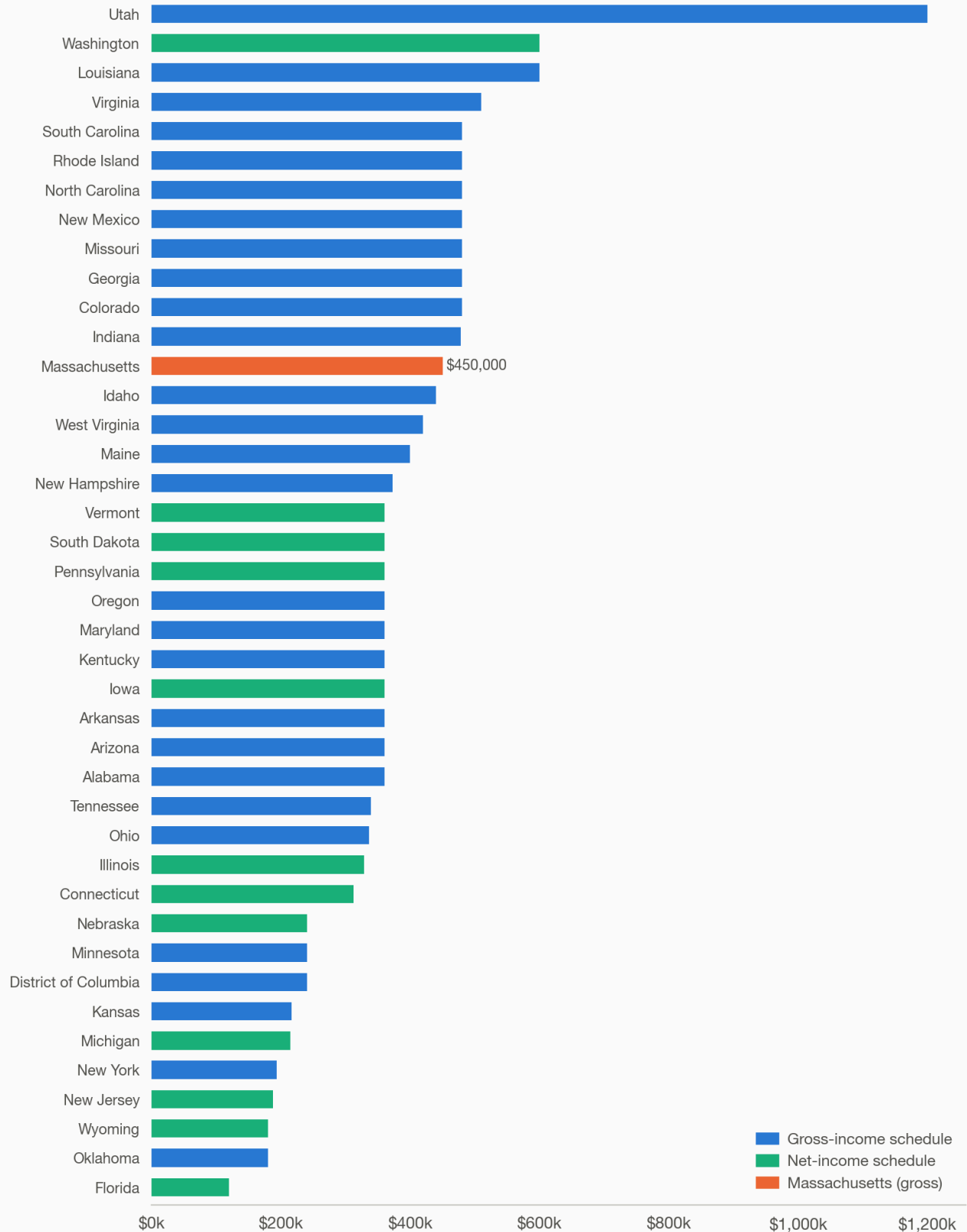
E16. Massachusetts's presumptive formula runs to \$450,000; 12 schedules run higher, 28 stop lower. Forty-one jurisdictions have a combined-income schedule that stops at a stated figure; the median is \$360,000, seven stop at exactly \$40,000 a month, and Utah's runs to \$1.2 million. Ten jurisdictions use a percentage-of-obligor, Melson or open formula with no combined ceiling. Net-income ceilings are not dollar-for-dollar comparable with gross ones. This figure corrects an earlier comparison against nine benchmark states, in which Massachusetts appeared second; of the 41 jurisdictions with a stated combined-income ceiling it is thirteenth.

Massachusetts's presumptive formula runs to \$450,000; 12 schedules run higher, 28 stop lower

Combined income at which each state's presumptive schedule ends; above it, support is discretionary.

RANKED 41 combined-income schedules (29 gross, 11 net, 1 other)

NOT RANKED 10: percentage-of-obligor, Melson or open formula · BASIS Annual; monthly ×12, weekly ×52



Notes: Median of the 41: \$360,000; 7 stop at exactly \$40,000 a month. Net-income ceilings are not dollar-for-dollar

comparable with gross ones. Not ranked: Alaska, California, Delaware, Hawaii, Mississippi, Montana, Nevada, North Dakota, Texas, Wisconsin.

Source: data/fifty-state/ceilings-2026-09-06.json (verified-run profiles completed from local primary documents)

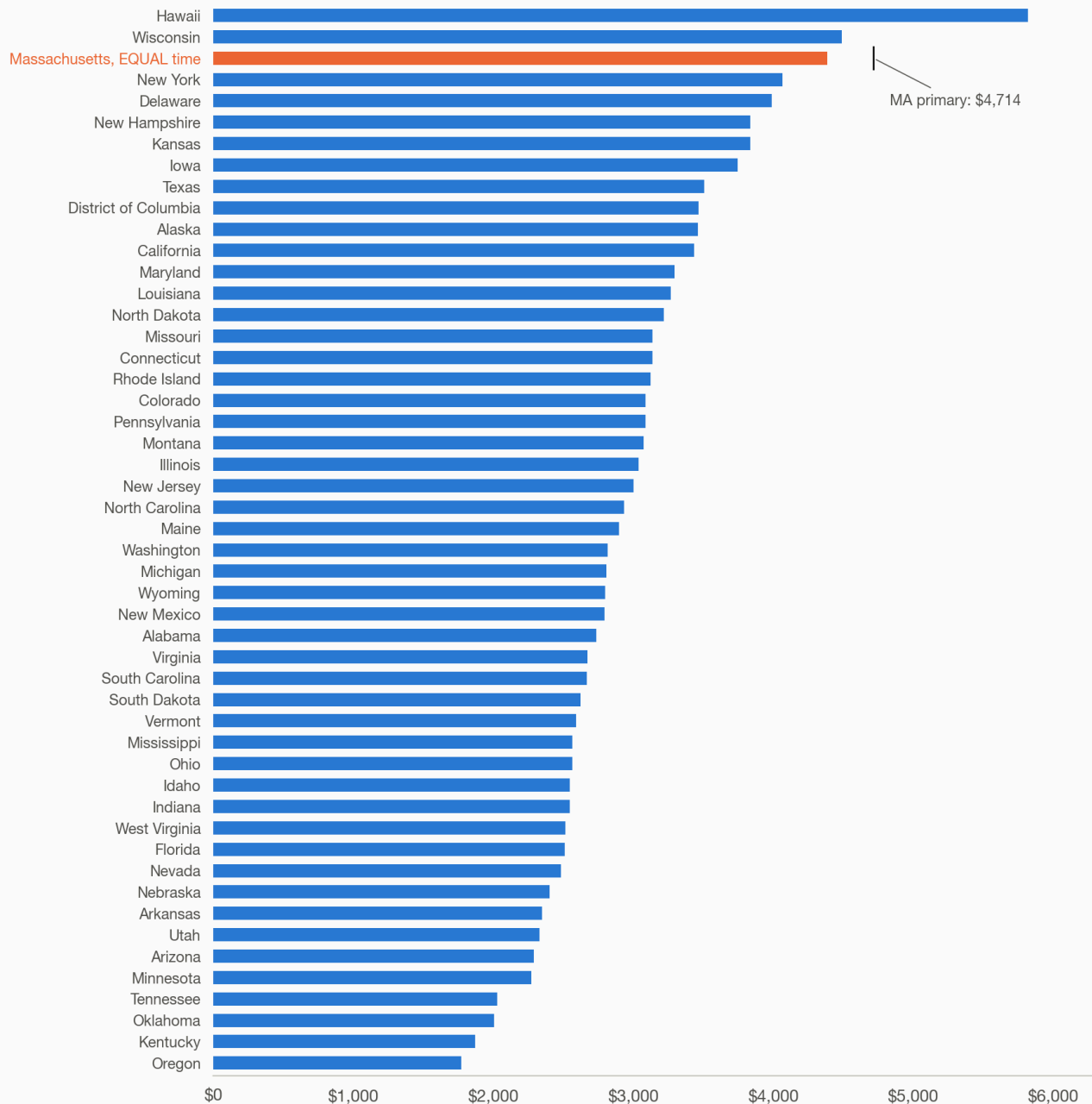
E17. Massachusetts charges more for joint custody than 47 states charge when the recipient has primary custody. The same fact pattern as E11 and E12, with Massachusetts computed under Box 1, the children half the time with each parent, and every other jurisdiction computed with the children primarily with the lower earner. Only Hawaii and Wisconsin, of the 49 other ranked jurisdictions, order more at primary custody than Massachusetts does at equal time; Massachusetts's own primary-custody order, \$4,714, is marked for scale. The Commonwealth's consultant attributes the higher amounts to the cost of living, which bears on both households in the order; whatever it explains about the level, it does not explain why an arrangement that gives each parent half the children's time is priced here where sole primary custody is priced almost everywhere else.

Only Wisconsin and Hawaii order more than Massachusetts does at equal time

Massachusetts's equal-time order against every other state's primary-custody order.

CUSTODY MA equal time (Box 1) vs others primary · CHILDREN 3 · CHILD CARE None (base support)

INCOMES \$201,000 / \$29,640



Notes: Scaled to match E12 so the two can be read side by side. Georgia held out; one fact pattern only.

Source: data/fifty-state/tier-50-2026-09-05.json

The gross-versus-net question

E27. Five reviews have taken up gross versus net. None changed it. Every Massachusetts guidelines review or economic review in the corpus, seven cycles from 2002 to 2025, marked filled where that cycle's own report or economic review states the gross-versus-net income basis question was raised and declined, and hollow where the corpus holds no record of it being raised at all. Each

filled year carries a verbatim quote from that cycle's own document, with a page number, in docs/2026-09-08-prior-taskforce-reports.md: the 2009 Task Force found "little support ... for basing orders on net income"; the 2012 Task Force "declines to recommend this change" on grounds of simplicity; 2017, 2021 and 2025 are statements to the same effect in the Commonwealth's consultant's economic review for that cycle, not a task force vote, so the chart and this caption say "reviews," never "task forces declined." The 2002 and 2006 cycles say nothing on the question in their own text, so they are marked no record rather than deferred. A separate, related question, how the guidelines and an alimony order interact for tax purposes, was also deferred in 2017 but is not counted here, because it is not the gross-versus-net income-basis question.

Five reviews have taken up gross versus net. None changed it

Massachusetts guidelines reviews since 2002. Filled: that cycle's own report or economic review takes up the question and recommends no change.

QUESTION Gross vs. net income basis (not the separate alimony/tax question)

CORPUS Every task force report and economic review, 2001 to 2025 · CYCLES 7 reviews, 2002 to 2025



Notes: Each filled year has a verbatim quote from that cycle's own document, listed with page numbers in the source file. The 2002 and 2006 cycles say nothing on the question, so they are marked no record rather than declined.

Source: data/deferrals-gross-vs-net.json (data/extracted/taskforce/)

Sources

Every figure names its script and data on its own Source line. Models: model/worksheet.py (CJ-D 304, matched line for line to the form's own calculation scripts on six scenarios), model/net_position.py (TY2026 federal and Massachusetts tax), model/box1_fix.py, model/childcare_post_transfer.py, model/marginal_retention.py. Fifty-one-jurisdiction data: data/fifty-state/tier-50-2026-09-05.json, credit-at-122-2026-09-05.json, ceilings-2026-09-06.json. Gross-versus-net deferral data, built by grepping data/extracted/*.flow.txt:

data/deferrals-gross-vs-net.json. Each figure's plotted values: the CSV of the same name in output/charts/.